

ULYANA DUPLETSA - CV

PhD in Astroparticle Physics

ABOUT ME

I am an astrophysicist focused on studying gravitational waves from compact binaries, namely binary black hole and binary neutron star systems. My research involves both current detectors, such as the LIGO-Virgo-KAGRA Collaboration, and future instruments like the Einstein Telescope. I am interested in data analysis methods, from Fisher matrix forecasts, to Bayesian parameter estimation, including enhancements from machine learning, applied to cosmology and multi-messenger astrophysics.

CONTACT INFO

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GitHub  <https://github.com/u-dupletsa>
Website <https://ulyanadupletsa.github.io/>

CURRENT POSITION

Postdoctoral Research Fellow Mar 2025-current
Marietta Blau Institute (MBI), Austrian Academy of Sciences — Vienna, Austria

EDUCATION

PhD in Astroparticle Physics Nov 2020-Feb 2025
Gran Sasso Science Institute — L'Aquila, Italy

- PhD Thesis on “Cosmic Chirps: Extracting Cosmological and Astrophysical Information from Compact Binary Mergers”, defended cum Laude on 16th May 2025
- Supervisor: Prof. Jan Harms; Co-Supervisors: Prof. Marica Branchesi, Dr. Riccardo Murgia

Master's degree in Theoretical Physics - 110/110 cum Laude 2017-2019
University of Milano Bicocca — Milan, Italy

- Thesis on “Thermodynamic Aspects of AdS_4 Black Holes in $N = 2$ Gauged Supergravity” supervised by Prof. Alberto Zaffaroni

Bachelor's Degree in Physics - 110/110 cum Laude 2013-2017
The University of Milano Bicocca — Milan, Italy

- Thesis on “Supermassive Binary Black Holes and their Dynamics in Galactic Nuclei” supervised by Prof. Monica Colpi

PHD SCHOOLS

IWR School on Machine Learning for Fundamental Physics [\[link\]](#) 15 - 19 Sep, 2025
Heidelberg, Germany

Computing and High Performance Computing in Astronomy and Astrophysics [\[link\]](#) 24 Jun-5 Jul, 2024
Bologna, Italy

International School for Subnuclear Physics 2024 - News from the four interactions [\[link\]](#) 14-23 Jun, 2024
“Ettore Majorana Foundation” - Erice, Italy

- Winner of *New Talents: Giazotto prize* 
- Prize for *Best Presentation in Theoretical Physics* 

Scientific Communication in Astronomy School [\[link\]](#) 2-6 Oct, 2023
Bertinoro, Italy

- Winner of the school prize (3rd place) 

First EuCapt School in Cosmology [\[link\]](#) 18-22 Sep, 2023
Valencia, Spain

Amaldi Research Center Summer School [\[link\]](#) 5-9 Sep, 2022
Paestum, Italy

The Onassis Foundation Science Lectures in Physics: Gravitational Waves [\[link\]](#) 25-29 Jul, 2022
Heraklion, Crete

INVITED TALKS & SEMINARS

Multiple Systems across Scales: First early-career researcher workshop of CRC1831 “Multiplicity of Stars and Planets” [\[link\]](#) 27-28 Apr, 2026
Heidelberg, Germany

- Talk on: “Standard siren cosmology: current status and prospects for the Einstein Telescope”

Seminar at Sapienza [link] Rome, Italy	4 Mar, 2026
• Seminar on: “The present and future of GW cosmology”	
Second International Conference on the Physics of the Two Infinities [link] Tokyo, Japan	17-21 Nov, 2025
• Talk on behalf of the ET Collaboration on: “The Einstein Telescope: challenges and opportunities of a third-generation detector”	
1st Sardinian GW science mini-workshop [link] Cagliari, Italy	22-26 Sep, 2025
• Talks and discussions on: “Cosmology and Black Hole Physics with GWs and 21cm intensity mapping”	
LXVI Congresso nazionale della Società Astronomica Italiana [link] Firenze, Italy	3-6 Jun, 2025
• Talk on: “Le sirene delle onde gravitazionali: calibrazione di un nuovo metodo per stimare H_0 ”	
GraSP24 - Gravity Shape Pisa 2024 [link] Pisa, Italy	23-25 Oct, 2024
• Talk on: “Forecasting detection and parameter estimation capabilities for the Einstein Telescope”	
Seminar at MBI [link] Vienna, Austria	10 Sep, 2024
• Seminar on: “Forecasting detection and parameter estimation capabilities for the Einstein Telescope”	
The Seventeenth Marcel Grossmann Meeting [link] Pescara, Italy	7-12 Jul, 2024
• Talk on “How to do multi-messenger forecasts in the Einstein Telescope era: addressing present and future challenges”	
Seminar at Cagliari University [link] Sardinia, Italy	23 Apr, 2024
• Seminar on: “ET forecasts on cosmological growth of BH masses”	
ET: Scienza e Tecnologia in Italia [link] Assisi, Italy	20-23 Feb, 2024
• Panelist in “Science and data analysis challenges”	
GWADW - Gravitational Wave Advanced Detector Workshop [link] Isola d’Elba, Italy	22-26 May, 2023
• Talk on: “Forecasting the Detection and Parameter Estimation Capabilities for different ET Designs”	
XIII ET Symposium [link] Cagliari, Italy	8-12 May, 2023
• Talk on: “Fisher Information Matrix for ET Forecasts: How Informative is it?”	

CONFERENCES & WORKSHOPS

LIGO-Virgo-KAGRA Collaboration meeting [link] Pisa, Italy	9-13 Mar, 2026
Einstein Telescope science workshop for early career researchers [link] Sapienza University - Rome, Italy (Co-organizer of the workshop)	18-20 Feb, 2026
2025 SKAO General Science Meeting [link] Görlitz, Germany	16-20 Jun, 2025
• Talk on: “Mapping the Universe’s Expansion with Gravitational Waves and Neutral Hydrogen”	
XV ET Symposium [link] Bologna, Italy	26-30 May, 2025
• <u>Plenary</u> talk (shared with Mikhail Korobko) on: “OSB work for the ETO Task Force”	
LIGO-Virgo-KAGRA Collaboration Meeting [link] <i>[online]</i>	24-27 Mar, 2023
• Talk (shared with Gregoire Pierra) on: “Latest results from the GWTC-4 cosmology paper”	
1st GSSI-JHU MAECI Meeting on Gravitational Wave Astrophysics [link] L’Aquila, Italy	16-18 Dec, 2024
• Talk on: “Dark sirens cosmology”	
AHEAD2024 2nd General Meeting [link] Rome, Italy	26-27 Nov, 2024
• Talk on: “Parameter estimation catalogs for binary neutron star mergers detected with next-generation gravitational wave detectors”	

LIGO-Virgo-KAGRA Collaboration Meeting [link] <i>Barcelona, Spain</i>	23-26 Sept, 2024
• Talk (shared with Suvodip Mukherjee) on: “Updates on Mock Data Challenge - Analysis with icarogw”	
ModIC 2024 - Model-Independent Cosmology with gravitational waves, large-scale structure, and high-energy surveys [link] <i>IFPU Trieste, Italy</i>	13-17 May, 2024
• Talk on: “Dark Sirens Cosmology with neutral hydrogen intensity mapping”	
XIV ET Symposium [link] <i>Maastricht, Netherlands</i>	6-10 May, 2024
• Talk on: “Enhancing Fisher Matrix Results with Physically Motivated Priors”	
Second Cosmic Explorer Symposium [link] <i>[online]</i>	23-25 Apr, 2024
• Short talk (1 minute 1 slide) on: “Enhancing Fisher Matrix Results with Physically Motivated Priors”	
Linking Advances in our Understanding of Theoretical Astrophysics and Relativity to Observations (LAUTARO) [link] <i>Milano-Bicocca, Italy</i>	17-19 Apr, 2024
• Talk on: “Enhancing Fisher Matrix Results with Physically Motivated Priors”	
LIGO-Virgo-KAGRA Collaboration Meeting [link] <i>[online]</i>	11-14 Mar, 2023
• Talk (shared with Suvodip Mukherjee) on: “MDC - Status and Prospects”	
LIGO-Virgo-KAGRA Collaboration Meeting [link] <i>Toyama, Japan</i>	11-14 Sept, 2023
• Talk (shared with Suvodip Mukherjee) on: “Mock Data Challenge - Analysis with icarogw”	
Amaldi 15 [link] <i>[online]</i>	17-21 Jul, 2023
• Talk on: “Forecasting detection and parameter estimation capabilities for different ET designs in a multi-messenger context”	
Gravitational-wave populations: what’s next? Workshop [link] <i>Milan, Italy</i>	10-14 Jul, 2023
• Short talk on: “GWFish - a Fisher Matrix Software”	
OSB - Population Studies, Monthly meeting <i>[online]</i>	14 Mar, 2023
• Talk on: “GWFish - a Fisher Matrix Analysis Software”	
OSB - Common Tools, Monthly meeting <i>[online]</i>	10 Mar, 2023
• Talk on: “Fisher Matrix Analysis - adding Priors”	
LIGO-Virgo-KAGRA Collaboration Meeting <i>Cardiff</i>	12-16 Sep, 2022
• Poster on: “GWFish - A simulation software to evaluate parameter-estimation capabilities of gravitational-wave detector networks”	
XII ET Symposium - The birth of the ET Collaboration [link] <i>Budapest, Hungary</i>	7-8 Jun, 2022

VISITINGS

GSSI <i>L’Aquila, Italy</i>	29 Apr-16 May, 2026
• Visiting on ACME funds (link) , collaboration on multi-messenger astrophysics and cosmology with the Einstein Telescope	
Sapienza University <i>Rome, Italy</i>	3-8 Mar, 2026
• Visiting Simone Mastrogiovanni’s group, and seminar on: “The present and future of GW cosmology” (link)	
GSSI <i>L’Aquila, Italy</i>	18-23 May, 2025
• Discussion and collaboration for the ET Task Force documents preparation	
University of Cagliari <i>Cagliari, Italy</i>	22-26 Apr, 2024
• Seminar on “ET forecasts on cosmological growth of BH masses”	
Johns Hopkins University <i>Baltimore, United States</i>	Oct-Nov, 2023
• Collaboration with Emanuele Berti’s group (mainly with Dr. Ken Ng) as part of the exchange program MAECI MUR organized by Andrea Maselli, resulted in the publication of Dupletsa et al. 2025	

PUBLICATION LIST		
	A full list of publications is available below.	
REVIEWER		
	Referee for the Astrophysical Journal (ApJ), internal reviewer in LVK	
LECTURES & TUTORIALS		
	<i>PhD school on “Gravitational Waves: from theory to detection”</i> [link]	7-18 Jul, 2025
	<i>Vienna, Austria</i>	
	Hands-on session on “Metrics and tools for assessing the performance of future gravitational wave detectors” (GitHub )	
	<i>“Gravitational Waves and their detection” master course at the Technical University of Vienna</i>	4 Jun, 2025
	<i>Vienna, Austria</i>	
	One lecture on “Gravitational wave cosmology” [2 hours] as part of the course held by Gianluca Inguglia	
	ModIC 2024 - Model-Independent Cosmology with gravitational waves, large-scale structure, and high-energy surveys [link]	13-17 May, 2024
	<i>IFPU Trieste, Italy</i>	
	Hands-on session on GWFish with and without priors (GitHub )	
ORGANIZATION OF SCIENTIFIC EVENTS	ET: Scienza e Tecnologia in Italia [link]	20-23 Feb, 2024
	<i>Assisi, Italy</i>	
	School lecture (in collaboration with Jacopo Tissino) on “<i>GWFish: Simulation of gravitational-wave detector networks with Fisher-matrix PE</i>” (GitHub )	
	XIII ET Symposium [link]	8-12 May, 2023
	<i>Cagliari, Italy</i>	
	Tutorial session on “<i>GWFish - a Fisher Matrix Analysis Software</i>” (GitHub )	
	Co-organizer of the MBI Colloquia [link]	Jan 2026- <i>current</i>
	<i>Vienna, Austria</i>	
	Next generation gravitational wave observations: mini-colloquia [link]	21 Seb, 2026
ROLE IN LVK		
	LIGO-Virgo-KAGRA (LVK) Collaboration	
	<i>CBC Cosmology Group member</i>	
	Member of both analysis, with the <code>icarogw</code> pipeline, and paper writing teams for the Mock Data Challenge short-author Collaboration paper	
	Member of both the analysis, with the <code>icarogw</code> pipeline, and the writing teams for the O4a Cosmology Collaboration paper; member of the analysis team for the O4b Cosmology Collaboration paper (<i>in preparation</i>)	
	Reviewer for the GWTC-5 Rates & Population Collaboration paper scripts	
	Einstein Telescope (ET) Collaboration	
	<i>Observational Science Board (OSB) member</i>	
	Member of the following divisions: Common Tools, Multi-Messenger Observations and Cosmology	
ROLE IN ET	Comparison and testing of Fisher matrix codes within the <i>Common Tools</i> division	
	Key contribution to the Cost and Benefit Analysis (CoBA) work published in Branchesi et al. 2023: forecast analysis of binary neutron star population with <code>GWFish</code>, cross-checks of the binary black hole population results	

- Key contribution to the [Blue Book](#), a more-detailed scientific follow-up of [Branchesi et al. 2023](#): part of the editorial and writing team of the *Common Tools* division chapter, and part of the writing team for the *Multi-Messenger Observations* division chapter, OSB liaison, together with Francesco Iacovelli and Archisman Ghosh, for the ETO Task Force detector layouts study: coordination and analyses work to assess the scientific risks of different sensitivity curves for the ET project. This comprised three workshops as detailed below:

ETO Task Force Meeting (I) [\[link\]](#)

18-20 Feb 2025, *[online]*

- Talks and discussions on: “Noise budget and science case of ET”

ETO Task Force Meeting (II) [\[link\]](#)

18-20 Mar 2025, *Nikhef, Amsterdam*

- Talks and discussions on: “Noise budget and science case of ET”

ETO Task Force Meeting (III) [\[link\]](#)

5-7 May 2025, *CERN, Switzerland*

- Talks and discussions on: “Noise budget and science case of ET”

- Coordinator (together with Francesco Iacovelli) of the science paper on the results from the ETO task force optical layout. Currently under review in the ET Collaboration.

FUNDS & GRANTS

ACME funds [\(link\)](#)

Oct 2025

Second TNA call to Centers of Expertise

- Selected for a two-week visiting program at GSSI [\(link\)](#)

APPHEAL Project - mini grant

2025

Advanced Platform for Processing Heavy Elements Libraries

- Principal Investigator: Dr. Eleonora Loffredo
- Description: 20k project funded by the Italian National Institute for Astrophysics
- Role: Co-Investigator

SKILLS

Programming languages

- Python, Bash (advanced), C/C++, Mathematica (intermediate), MATLAB, SQL (basic)

Software and Tools

- L^AT_EX, git, condor, slurm

Languages

- Italian (native), Ukrainian (native), English (advanced)

OUTREACH ACTIVITIES

Lange Nacht der Forschung — *Vienna, Austria*

- Outreach event organized by the Austrian Academy of Sciences

SHARPER — *L'Aquila, Italy*

- Yearly participation on behalf of GSSI to the outreach activities of the European Researchers' Night (24 September 2021, 30 September 2022, 29 September 2023 and 27 September 2024)

IDeAS — *L'Aquila, Italy*

- member of IDeAS (IDeAS is partner of Museo Nivola of Organi, GSSI, GSSI Alumni, INFN Cagliari, University of Cagliari, and INAF-OAC, in the realization of the project “*On the edge of the eyelid*”, winner of *Strategia Fotografica 2024* from *Ministero della Cultura*)
- Public seminar on “*Seeing the Unseen*”, in collaboration with Matteo Schulz and Andrea Cozzumbo, on 30th January 2025

Scienza in Bidda — *Jerzu, Sardinia*

- Event organized by Dr. Riccardo Murgia in 2023, in correspondence with the ET Symposium in Cagliari; part of the outreach project to talk about the relevance of the Einstein Telescope in Sardinia

Cosmic Explorers — *L'Aquila, Italy*

- Series of lectures by Prof. Marica Branchesi to elementary school kids, contribution to organization of activities and experiments with the kids; three days in March, April, and June 2023

PhTea Talks — *L'Aquila, Italy*

- Co-organizer and co-founder of informal talks for PhD students at GSSI
- Outreach talk: “*A Hitchhiker’s Guide to hush annoying Flat Earthers*”, March 2023
- Outreach talk: “*Sip back and relax: it’s tea time!*”, April 2024

Premio Asimov — *L'Aquila, Italy*

- Project organized by Prof. Francesco Vissani; part of the reviewing committee for the years 2021, 2022, 2023, 2024, 2025, 2026

PUBLICATION LIST

Ulyana Dupletsa

USEFUL LINKS

ORCID	0000-0003-2766-247X
Scopus	authorID:57238177800

PUBLICATION LIST

1. "Radio sirens: inferring H_0 with binary black holes and neutral hydrogen in the era of the Einstein Telescope and the SKA Observatory"
U. Dupletsa, S. Mastrogiovanni, M. Spinelli, T. Ronconi, M. Schulz, R. Murgia, J. Harms, T. Baker, M. Calabrese, C. Carbone, S. Cunningham, I. Harrison, K. Leyde, D. Nanadoumgar-Lacroze; *submitted to MNRAS*, arXiv:2605.12606 [astro-ph.CO] [astro-ph.HE]
2. "Probing Cosmic Expansion and Early Universe with Einstein Telescope"
A. Ricciardone et al. incl. **U. Dupletsa**; *ESO Expanding Horizon White Paper*, arXiv:2601.06017 [astro-ph.CO] [gr-qc]
3. "Comparing next-generation detector configurations for high-redshift gravitational wave sources with neural posterior estimation"
F. Santoliquido, J. Tissino, **U. Dupletsa**, M. Branchesi, J. Harms; *Astronomy & Astrophysics*, 708 (2026) A175, arXiv:2512.20699 [gr-qc] [astro-ph.HE]
4. "Towards an anomaly detection pipeline for gravitational waves at the Einstein Telescope"
G. Inguglia, H. Haigh, K. Vitulova, **U. Dupletsa**; *Physical Letters B*, 874 (2026), arXiv:2511.13154 [gr-qc] [astro-ph.IM]
5. "Multi-messenger observations of binary neutron star mergers: synergies between the next generation gravitational wave interferometers and wide-field, high-multiplex spectroscopic facilities"
S. Bisero, D. Vergani, E. Loffredo, M. Branchesi, H. Nandini, **U. Dupletsa**, R. I. Anderson; *Astronomy & Astrophysics*, 705 (2026) A54, arXiv:2507.02055 [astro-ph.HE]
6. "Fast and accurate parameter estimation of high-redshift sources with the Einstein Telescope"
F. Santoliquido, J. Tissino, **U. Dupletsa**, M. Branchesi, J. Harms, M. Arca Sedda, M. Dax, A. Kofler, S. R. Green, N. Gupta, I. M. Romero-Shaw, E. Berti; *PRD*, 112 (2025) 10, arXiv:2504.21087 [astro-ph.HE] [astro-ph.IM] [gr-qc]
7. "The Science of the Einstein Telescope"
A. Abac et al. incl. **U. Dupletsa**; *submitted to JCAP*, arXiv:2503.12263 [gr-qc]
8. "Blinded Mock Data Challenge for Gravitational-Wave Cosmology-I: Assessing the Robustness of Methods Using Binary Black Holes Mass Spectrum"
A. Agarwal, **U. Dupletsa**, K. Leyde, S. Mukherjee, B. Revenu, J. E. Rivera, A. E. Romano, M. R. Sah, S. Vallejo-Pena, A. Avendano, F. Beirnaert, G. Dalya, M. C. Espitia, C. Karathanasis, S. Moreno-Gonzalez, L. Quiceno, F. Stachurski, J. Garcia-Bellido, R. Gray, N. Tamanini, C. Turski; *The Astrophysical Journal*, Vol. 987 (2025), DOI:10.3847/1538-4357/adda3a, arXiv:2412.14244 [astro-ph.CO], GitHub 
9. "Model-independent cosmology with joint observations of gravitational waves and γ -ray bursts"
A. Cozzumbo, **U. Dupletsa**, R. Cald ron, R. Murgia, G. Oganesyan, M. Branchesi; *JCAP*, Vol. 05 (2025) 021, DOI:10.1088/1475-7516/2025/05/021, arXiv:2411.02490 [gr-qc]
10. "Parameter estimation catalogs for binary neutron star mergers detected with next-generation gravitational wave detectors", *Zenodo catalogs*
U. Dupletsa, F. Santoliquido, M. Branchesi, E. Loffredo; Sept. 2024,
11. "Prospects for optical detections from binary neutron star mergers with the next-generation multi-messenger observatories"
E. Loffredo, N. Hazra, **U. Dupletsa**, M. Branchesi, S. Ronchini, F. Santoliquido, A. Perego, B. Banerjee, S. Bisero, G. Ricigliano, S. Vergani, I. Andreoni, M. Cantiello, J. Harms, M. Mapelli, G. Oganesyan; *Astronomy & Astrophysics*, Vol. 697 (2025) A36, DOI:10.1051/0004-6361/202452863, arXiv:2411.02342 [astro-ph.HE], public catalogs of data available on Zenodo
12. "Validating Prior-informed Fisher-matrix Analyses against GWTC Data"
U. Dupletsa, J. Harms, Ken K. Y. Ng, J. Tissino, F. Santoliquido, A. Cozzumbo; *Physical Review D*, Vol. 111 (2025), 2, 024036, DOI:10.1103/PhysRevD.111.024036, arXiv:2404.16103 [gr-qc], GitHub 
13. "Classifying binary black holes from Population III stars with the Einstein Telescope: a machine-learning approach"
F. Santoliquido, **U. Dupletsa**, J. Tissino, M. Branchesi, F. Iacovelli, G. Iorio, M. Mapelli, D. Gerosa, J. Harms and M. Pasquato ; *Astronomy & Astrophysics*, Vol. 690 (2024), A362, DOI:10.1051/0004-6361/202450381, arXiv:2404.10048 [astro-ph.HE]

14. "The Wide-field Spectroscopic Telescope (WST) Science White Paper"
V. Mainieri et al. incl. **U. Dupletsa**; 2024, [arXiv:2403.05398](#) [astro-ph.IM]
15. "Phenomenological models of Cosmic Ray transport in Galaxies"
C. Evoli and **U. Dupletsa**; 2023, in *Proceedings of the International School of Physics "Enrico Fermi"*, Vol. 208: Foundations of Cosmic Ray Astrophysics, [arXiv:2309.00298](#) [astro-ph.HE]
16. "Science with the Einstein Telescope: a comparison of different designs"
M. Branchesi, M. Maggiore et al. incl. **U. Dupletsa**; *JCAP*, Vol. 07 (2023) 068, DOI:10.1088/1475-7516/2023/07/068, [arXiv:2303.15923](#) [gr-qc] [astro-ph.CO] [astro-ph.HE]
17. "Pre-merger alert to detect the very-high-energy prompt emission from binary neutron-star mergers: Einstein Telescope and Cherenkov Telescope Array synergy",
B. Banerjee, G. Oganesyan, M. Branchesi, **U. Dupletsa**, F. Aharonian, F. Brighenti, B. Goncharov, J. Harms, M. Mapelli, S. Ronchini, F. Santoliquido
Astronomy & Astrophysics, Vol. 678, DOI:10.1051/0004-6361/202345850, [arXiv:2212.14007](#) [astro-ph.HE]
18. "Measuring properties of primordial black hole mergers at cosmological distances: effect of higher order modes in gravitational waves",
K. K. Y. Ng, B. Goncharov, S. Chen, S. Borhanian, **U. Dupletsa**, G. Franciolini, M. Branchesi, J. Harms, M. Maggiore, A. Riotto, B. S. Sathyaprakash, S. Vitale, *Physical Review D*, Vol. 107, 024041, DOI:10.1103/PhysRevD.107.024041, [arXiv:2210.03132](#) [astro-ph.HE] [gr-qc]
19. "GWFish: A simulation software to evaluate parameter-estimation capabilities of gravitational-wave detector networks"
U. Dupletsa, J. Harms, B. Banerjee, M. Branchesi, B. Goncharov, A. Maselli, A. C. S. Oliveira, S. Ronchini, J. Tisino; *Astronomy and Computing*, Vol. 42 (2023), DOI:10.1016/j.ascom.2022.100671, [arXiv:2205.02499](#) [gr-qc], [GitHub](#) 
20. "Perspectives for multi-messenger astronomy with the next generation of gravitational-wave detectors and high-energy satellites",
S. Ronchini, M. Branchesi, G. Oganesyan, B. Banerjee, **U. Dupletsa**, G. Ghirlanda, J. Harms, M. Mapelli, F. Santoliquido; *Astronomy & Astrophysics*, Vol. 665 (2022), A97, DOI:10.1051/0004-6361/202243705, [arXiv:2108.07276](#) [astro-ph.HE]
21. "On the single-event-based identification of primordial black hole mergers at cosmological distances",
K. K. Y. Ng, S. Chen, B. Goncharov, **U. Dupletsa**, S. Borhanian, M. Branchesi, J. Harms, M. Maggiore, B. S. Sathyaprakash, S. Vitale; *ApJL*, Vol. 931 (2022) L12, DOI 10.3847/2041-8213/ac6bea, [arXiv:2108.07276](#) [gr-qc], [hep-ph]

COLLABORATION PAPERS

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22. "GWTC-4.0: Constraints on the Cosmic Expansion Rate and Modified Gravitational-wave Propagation"
LIGO-Virgo-KAGRA Collaboration; [arXiv:2509.04348](#) [astro-ph.CO], accepted in *APJ Letters* (Member of analysis and paper writing teams)
 23. **25+ Collaboration Papers inside the LIGO-Virgo-KAGRA Collaboration** [see complete list [here](#)]